

Lecture Notes: Probability Recap and Bayesian Decision Theory

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0.1 Notation Summary

Symbol	Meaning	Example
Ω	Sample space	$\Omega = \{H, T\}$ for a coin toss
A, B	Events	$A = \{\text{Heads}\}$
$P(A)$	Probability of event A	$P(A) = 0.5$
A^c	Complement of event A	$A^c = \{\text{Tails}\}$
$A \cap B$	Intersection of events	$\{\text{Heads and Rain}\}$
$A B$	Event A given B	$P(\text{Cancer} \text{Test+})$

1 Probability Recap (Event-based)

1.1 Sample Space and Events

A *sample space* Ω is the set of all possible outcomes of a random experiment. For example, when tossing a fair coin twice:

$$\Omega = \{(H, H), (H, T), (T, H), (T, T)\}.$$

An *event* is any subset of Ω . For instance, define:

$$\begin{aligned} A &= \{\text{second toss is } H\} = \{(H, H), (T, H)\}, \\ B &= \{\text{both tosses are the same}\} = \{(H, H), (T, T)\}. \end{aligned}$$

1.2 Probability Measure

A probability measure P assigns each event $A \subseteq \Omega$ a number $P(A)$ such that:

- $0 \leq P(A) \leq 1$,
- $P(\Omega) = 1$,
- If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

Complement Rule: For any event A ,

$$P(A^c) = 1 - P(A)$$

This follows directly from the axioms since A and A^c partition Ω .

For our coin toss example (assuming fair coins and independence),

$$P((H, H)) = P((H, T)) = P((T, H)) = P((T, T)) = \frac{1}{4}.$$

1.3 Conditional Probability

If $P(B) \neq 0$, the *conditional probability* of A given B is defined by:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

This measures the chance that event A occurs provided that B has occurred.

Illustrative Example: Student Enrollment

Imagine a university where:

- 60% of the students are female.
- 62% of all students are enrolled in an advanced course.
- Among the female students, 70% are enrolled in that course.

Let:

- B be the event that a student is female.
- A be the event that a student is enrolled in the course.

Then, the conditional probability $P(A | B) = 70\%$ since 70% of the females are enrolled. Without conditioning, the enrollment rate is 62%. This example shows how conditioning on gender (additional information) can change the probability of enrollment.

The diagram in Figure 1 illustrates the student enrollment example. The blue circle represents female students (B , 60%), the red circle represents enrolled students (A , 62%), and their overlapping (orange) area represents female students who are enrolled (42%). The conditional probability is the ratio of the intersection to the female group:

$$P(A | B) = \frac{42\%}{60\%} \approx 70\%.$$

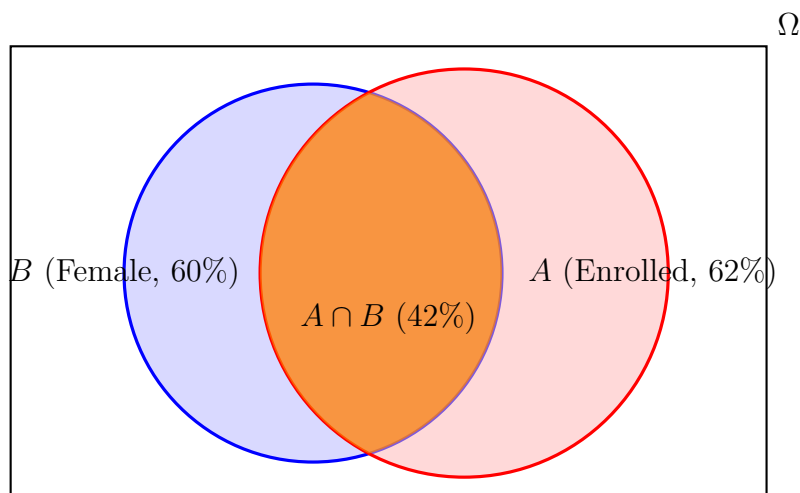


Figure 1: Venn Diagram for the student enrollment example, showing $P(A | B) = \frac{A \cap B}{B} \approx 70\%$.

Additional Example: Language Enrollment

In a school, suppose:

- 40% of students take Spanish.
- 30% of students take French.
- 15% of students take both Spanish and French.

Let:

- S : the event a student takes Spanish.
- F : the event a student takes French.

Then, the conditional probability that a student takes French given that they take Spanish is:

$$P(F | S) = \frac{P(F \cap S)}{P(S)} = \frac{0.15}{0.40} = 0.375.$$

Thus, if a student takes Spanish, there is a 37.5% chance they also take French.

1.4 Product Rule and Law of Total Probability

Product Rule (From the definition of conditional probability):

$$P(A \cap B) = P(A | B) P(B) = P(B | A) P(A).$$

Law of Total Probability: If $\{B_i\}$ is a partition of Ω , then

$$P(A) = \sum_i P(A | B_i) P(B_i).$$

Example: Let R be the event that it will rain today (R^c that it will not rain). Let J be the event you will wear a jacket (J^c that you will not wear a jacket). We know the following

- $P(R) = 0.8$ ($P(R^c) = 1 - P(R) = 0.2$).
- $P(J | R) = 0.9$ ($P(J^c | R) = 0.1$).
- $P(J | R^c) = 0.3$ ($P(J^c | R^c) = 0.7$).

Then the probability that you will wear a jacket today regardless is (total law of probability):

$$P(J) = P(J | R)P(R) + P(J | R^c)P(R^c).$$

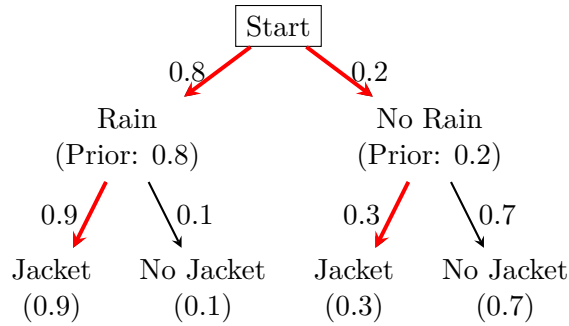


Figure 2: Probability tree for rain-jacket decision making. Highlighted paths show contributions to total jacket probability: $0.8 \times 0.9 + 0.2 \times 0.3 = 0.78$

1.5 Bayes' Rule and Its Philosophy

Using the conditional probability, we can write $P(A | B) = \frac{P(A \cap B)}{P(B)}$ and $P(B | A) = \frac{P(A \cap B)}{P(A)}$. From the second one, we can find $P(A \cap B) = P(B | A)P(A)$. If we substitute $P(A \cap B) = P(B | A)P(A)$ for the formula of $P(A | B)$, we get the Bayes' rule

$$P(A | B) = \frac{P(B | A) P(A)}{P(B)}$$

- **Prior** ($P(A)$): Initial belief about hypothesis A before seeing data
- **Likelihood** ($P(B|A)$): How probable data B is under hypothesis A
- **Evidence** ($P(B)$): Total probability of data across all hypotheses
- **Posterior** ($P(A|B)$): Updated belief after considering evidence B .

Interpretation

Bayesian reasoning formalizes the scientific method:

1. Start with prior beliefs (hypotheses)
2. Gather experimental evidence (data)
3. Update beliefs via Bayes' theorem
4. Repeat as new evidence emerges

This framework naturally handles iterative learning - today's posterior becomes tomorrow's prior. It also quantifies Occam's razor through the evidence term, which automatically penalizes overly complex hypotheses.

Medical Testing Example (Bayesian Perspective)

Suppose:

- The disease prevalence (prior) is $P(D) = 0.01$.
- The test is 90% sensitive, so $P(T | D) = 0.90$ (likelihood).
- The test has a 5% false positive rate, so $P(T | D^c) = 0.05$.

Then, the overall probability of a positive test (evidence) is:

$$P(T) = P(T | D)P(D) + P(T | D^c)P(D^c) = 0.90 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585.$$

Using Bayes' rule, the posterior probability of having the disease given a positive test is:

$$P(D | T) = \frac{0.90 \cdot 0.01}{0.0585} \approx 0.1538.$$

Thus, even with a positive test, the probability of having the disease is only about 15.4%—a result that underscores the importance of the low prior probability (base rate) in the Bayesian update. This is known as the *base rate fallacy* - even with an accurate test, low prior probabilities lead to surprising posterior results. Bayesian reasoning protects against this by rigorously combining all information sources.

1.6 Independence

Events A and B are independent if:

$$P(A \cap B) = P(A)P(B),$$

which also implies $P(A | B) = P(A)$ from the conditional probability definition.

2 Random Variables

2.1 Definition and Examples

A *random variable* X is a function that assigns a numerical value to each outcome in Ω . For example, if

$$\Omega = \{(H, H), (H, T), (T, H), (T, T)\},$$

we can define

$$X(\omega) = \text{number of Heads.}$$

Thus,

$$X((H, H)) = 2, \quad X((H, T)) = 1, \quad X((T, H)) = 1, \quad X((T, T)) = 0.$$

2.2 Discrete vs. Continuous Random Variables

- **Discrete:** X takes values from a countable set (e.g., number of heads, die outcomes).
- **Continuous:** X takes values from an interval (e.g., temperature, height).

3 Probability Rules for Random Variables (PMFs and PDFs)

These rules for probability distributions are analogous to those for events.

3.1 Discrete Random Variables: PMFs

For a discrete random variable $X : \Omega \rightarrow \mathbb{X}$ with probability mass function (pmf) $p_X(x)$:

$$P(X = x) = p_X(x), \quad \sum_{x \in \mathbb{X}} p_X(x) = 1.$$

Joint Distribution: For two discrete random variables X and Y :

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

Marginal Distribution:

$$p_X(x) = \sum_{y \in \mathbb{Y}} p_{X,Y}(x, y).$$

Conditional Distribution:

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Product Rule:

$$p_{X,Y}(x, y) = p_{X|Y}(x | y) p_Y(y) = p_{Y|X}(y | x) p_X(x).$$

Bayes' Rule:

$$p_{X|Y}(x | y) = \frac{p_{Y|X}(y | x) p_X(x)}{p_Y(y)}.$$

Independence: X and Y are independent if:

$$p_{X,Y}(x, y) = p_X(x) p_Y(y).$$

3.2 Continuous Random Variables: PDFs

For a continuous random variable $X : \Omega \rightarrow \mathbb{R}$ with probability density function (pdf) $p_X(x)$:

$$P(a \leq X \leq b) = \int_a^b p_X(x) dx, \quad \int_{-\infty}^{\infty} p_X(x) dx = 1, \quad p_X(x) \geq 0.$$

Note that $P(X = x) = 0$ for any specific x .

Similarly, for two continuous random variables X and Y :

Joint PDF:

$$p_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} P(X \leq x, Y \leq y).$$

Marginal PDF:

$$p_X(x) = \int p_{X,Y}(x, y) dy.$$

Conditional PDF:

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

Product Rule (Continuous):

$$p_{X,Y}(x, y) = p_{X|Y}(x | y) p_Y(y) = p_{Y|X}(y | x) p_X(x).$$

Bayes' Rule (Continuous):

$$p_{X|Y}(x | y) = \frac{p_{Y|X}(y | x) p_X(x)}{p_Y(y)}.$$

Independence (Continuous): X and Y are independent if:

$$p_{X,Y}(x, y) = p_X(x) p_Y(y).$$

4 Expectation

4.1 Definition

The *expectation* of a function $f(X)$ is defined as:

$$\mathbb{E}[f(X)] = \begin{cases} \sum_{x \in \mathbb{X}} p_X(x) f(x), & \text{if } X \text{ is discrete,} \\ \int_{-\infty}^{\infty} p_X(x) f(x) dx, & \text{if } X \text{ is continuous.} \end{cases}$$

In particular, $\mathbb{E}[X]$ is the mean of X .

4.2 Examples

Discrete Example: Fair Die

For a fair die with outcomes $1, 2, \dots, 6$ and $p_X(x) = \frac{1}{6}$,

$$\mathbb{E}[X] = \sum_{x=1}^6 \frac{1}{6} x = \frac{21}{6} = 3.5.$$

Continuous Example: Uniform Distribution on $[0,1]$

For X being distributed uniformly on the interval $[0, 1]$ ($X \sim U(0, 1)$) with $p_X(x) = 1$ for $0 \leq x \leq 1$,

$$\mathbb{E}[X] = \int_0^1 x dx = \frac{1}{2}.$$

4.3 Discrete Distributions

Bernoulli Distribution. Let X be a Bernoulli random variable with parameter p . Then,

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

The pmf can be written as

$$p_X(x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

If X is a Bernoulli random variable, then we usually write $X \sim \text{Ber}(p)$ to denote that it is distributed according to a Bernoulli distribution with parameter p . **Expected value:** $E[X] = p$.

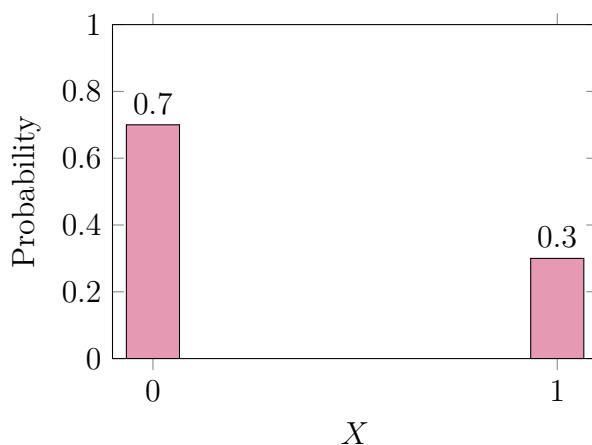


Figure 3: Bernoulli PMF (for $p = 0.3$).

Binomial Distribution. If X is the number of successes in n independent Bernoulli trials with probability p , then

$$p_X(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

If X is a Binomial random variable, then we usually write $X \sim \text{Binomial}(n, p)$ to denote that it is distributed according to a Binomial distribution with parameters n and p . **Expected value:** $E[X] = np$.

Poisson Distribution. For a Poisson random variable X with parameter λ ,

$$p_X(x) = P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

If X is a Poisson random variable, then we usually write $X \sim \text{Poisson}(\lambda)$ to denote that it is distributed according to a Poisson distribution with parameter λ . **Expected value:** $E[X] = \lambda$.

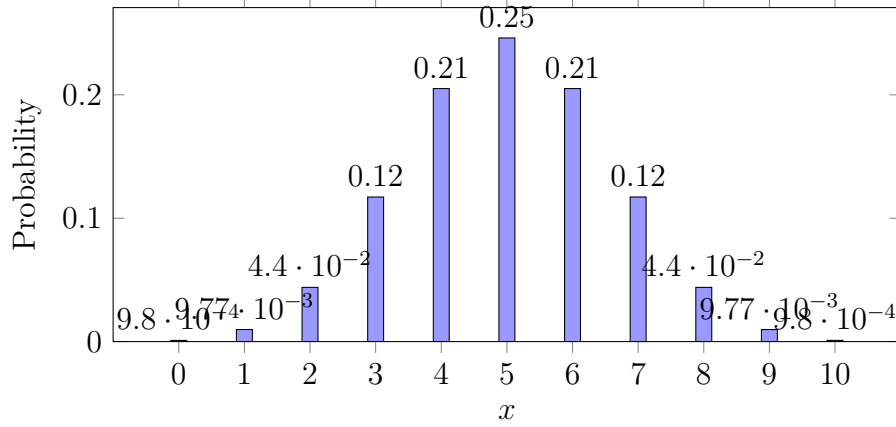


Figure 4: Binomial PMF (for $n = 10$, $p = 0.5$).

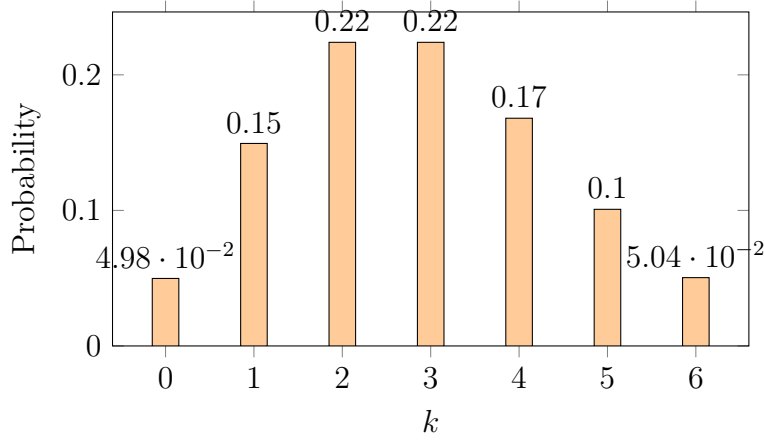


Figure 5: Poisson PMF (for $\lambda = 3$).

4.4 Continuous Distributions

Uniform Distribution. For a uniform random variable X on the interval $[a, b] \subset \mathbb{R}$, we have

$$p_X(x) = \frac{1}{b-a}, \quad x \in [a, b].$$

If X is a uniform random variable on the interval $[a, b]$, then we usually write $X \sim U(a, b)$ to denote it. **Expected value:** $E[X] = \frac{a+b}{2}$.

Normal Distribution. For a Gaussian (normal) random variable X with mean μ and variance σ^2 , we have

$$p_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

For a normal random variable, we usually write $X \sim \mathcal{N}(\mu, \sigma^2)$ to denote that X is normally distributed with parameters μ and σ^2 . **Expected value:** $E[X] = \mu$.

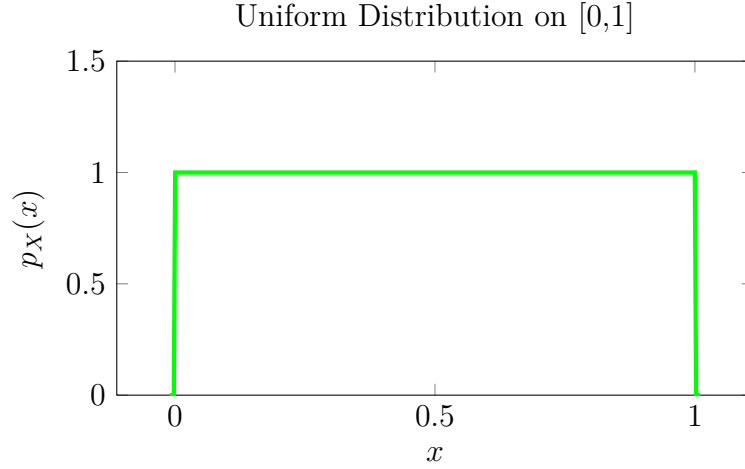


Figure 6: PDF of Uniform Distribution on $[0,1]$.

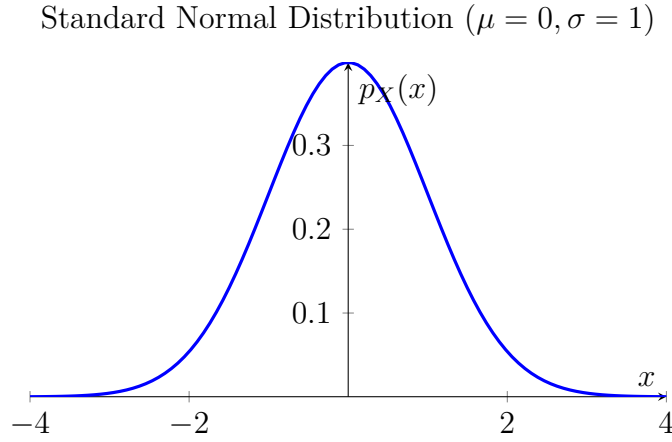


Figure 7: PDF of Standard Normal Distribution ($\mu = 0, \sigma = 1$).

Exponential Distribution. For an exponentially distributed random variable with rate parameter λ (mean $1/\lambda$), we have

$$p_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

For an exponentially distributed random variable, we write $X \sim \text{Exp}(\lambda)$. **Expected value:** $E[X] = \frac{1}{\lambda}$.

Laplace Distribution. For a Laplace random variable X with location parameter μ and scale parameter $b > 0$, the probability density function is given by

$$p_X(x) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right), \quad x \in \mathbb{R}.$$

If X is Laplace distributed, we write $X \sim \text{Laplace}(\mu, b)$. **Expected value:** $E[X] = \mu$.

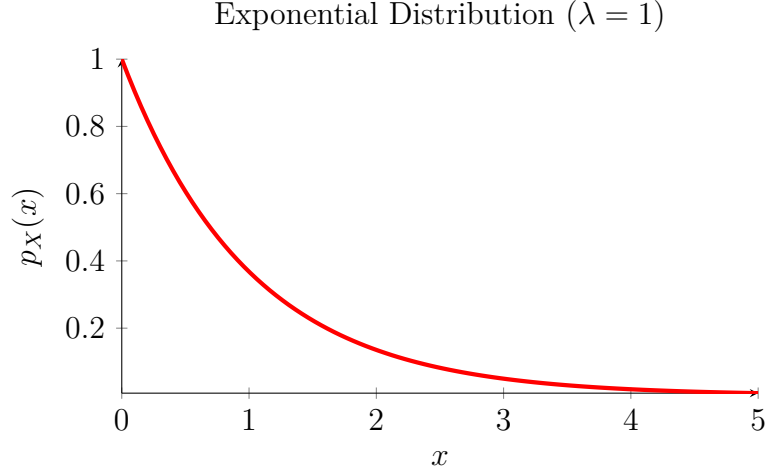


Figure 8: PDF of Exponential Distribution ($\lambda = 1$).

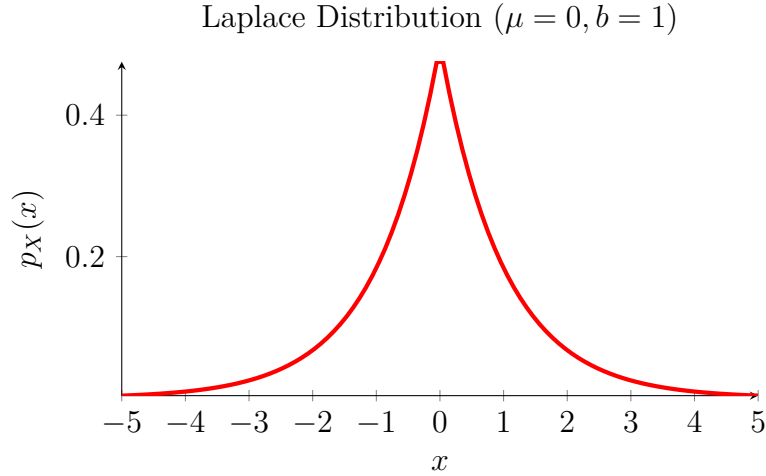


Figure 9: PDF of Laplace Distribution ($\mu = 0, b = 1$).

Multivariate Normal Distribution. For a random vector $\mathbf{X}$ in $\mathbb{R}^k$ that is distributed according to a multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, the probability density function is given by

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right), \quad \mathbf{x} \in \mathbb{R}^n.$$

If $\mathbf{X}$ is a multivariate normal random vector, we denote it by $\mathbf{X} \sim \mathcal{N}_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. **Expected value:** $E[\mathbf{X}] = \boldsymbol{\mu}$.

For $n = 2$ ($\mathbf{X} = [X_1, X_2]^\top$), the covariance matrix is

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}.$$

and $\rho \in [-1, 1]$ represent the correlation between X_1 and X_2 .

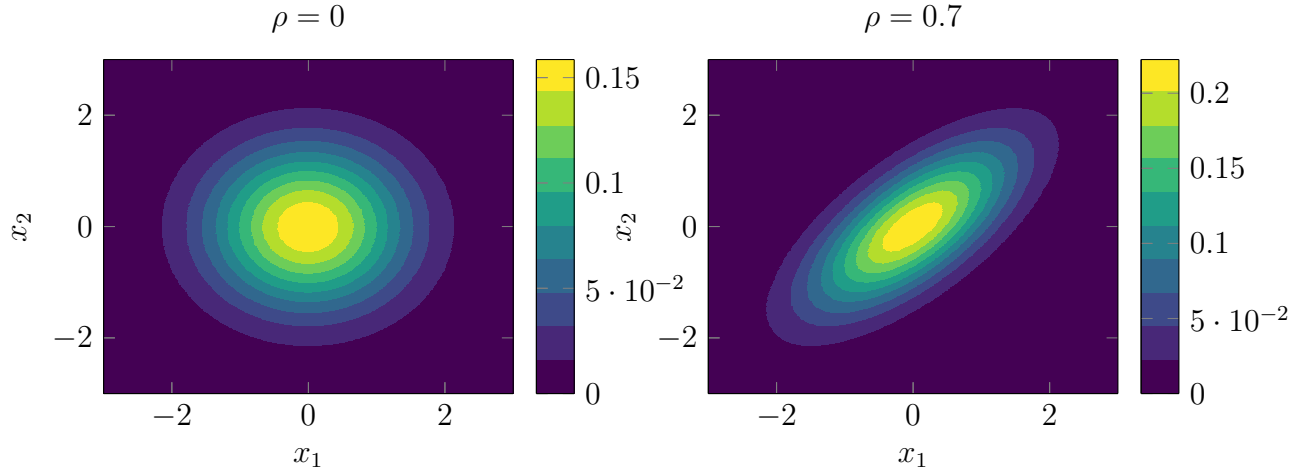


Figure 10: Comparison of 2D filled contour plots for the bivariate normal distribution: left, uncorrelated ($\rho = 0$) with circular contours; right, correlated ($\rho = 0.7$) with elliptical contours.

5 Bayesian Decision Theory

Bayesian Decision Theory provides a rigorous framework for making decisions under *uncertainty*, leveraging the probability distributions we have been discussing (e.g., $p_{Y|X}(y|x)$). In this section, we describe the key components, rules, and examples of Bayesian decision making using the exact notation and terminology as in the slides.

5.1 Overview

The main ingredients are:

- **Possible Outcomes:** Denote them by $Y \in \mathbb{Y}$ (e.g., $Y = 1, 2, \dots, K$ for K classes).
- **Measurements (Data):** Denote them by $X \in \mathbb{X}$, with a joint (or conditional) distribution

$$p_{X,Y}(x, y) = p_{Y|X}(y|x)p_X(x).$$

- **Actions (Decisions):** Denote by $a \in \mathcal{A}$ (e.g., choose label 1, choose label 2, ..., or “reject,” or “re-measure,” etc.).
- **Loss Function:** $L(y, a)$ quantifies how costly it is to take action a when the true outcome is y .

Goal:

Choose the action a^* that *minimizes* the expected loss (risk) given the observed data.

5.2 Posterior Distribution and Expected Loss

Once an observation $X = \mathbf{x}$ is made, we compute the posterior distribution over Y using Bayes' rule:

$$p_{Y|X}(y | x) = \frac{p_{X|Y}(x | y) p_Y(y)}{p_X(x)}.$$

The expected loss (conditional risk) for an action a is then given by

$$R(a | \mathbf{x}) = \sum_{y \in \mathbb{Y}} p_{Y|X}(y | \mathbf{x}) L(y, a) \quad (\text{discrete } Y),$$

or

$$R(a | \mathbf{x}) = \int_{y \in \mathbb{Y}} p_{Y|X}(y | \mathbf{x}) L(y, a) dy \quad (\text{continuous } Y).$$

The **Bayesian Decision Rule** is:

$$a^*(\mathbf{x}) = \underset{a \in \mathcal{A}}{\operatorname{argmin}} R(a | x).$$

5.3 0–1 Loss Case: Maximum A Posteriori Rule

For a classification problem with classes $\mathcal{C} = \{1, \dots, K\}$ and corresponding actions $\mathcal{A} = \{a_1, \dots, a_K\}$ (where a_i is the action “choose class c ” for $c \in \{1, \dots, K\}$), define the loss function as

$$L(c, a_i) = \begin{cases} 0, & \text{if } i = c \quad (\text{correct}), \\ 1, & \text{if } i \neq c \quad (\text{wrong}). \end{cases}$$

Then, the expected loss for choosing a_i given $X = x$ becomes

$$R(a_i | x) = \sum_{c=1}^K p_{C|X}(c | x) L(c, a_i) = 1 - p_{C|X}(i | x),$$

since

$$\sum_{c=1}^K p_{C|X}(c | x) = 1.$$

Thus, the decision rule simplifies to

$$a^*(\mathbf{x}) = a_{\underset{c \in \mathcal{C}}{\operatorname{argmax}} p_{C|X}(c | x)},$$

which is the well-known **Maximum A Posteriori (MAP)** rule.

5.4 Bayesian Decision-Making Depends on Proportionality

Recall that the posterior is proportional to the product of the likelihood and the prior:

$$\underbrace{p_{C|X}(c | x)}_{\text{Posterior}} \propto \underbrace{p_{X|C}(x | c)}_{\text{Likelihood}} \times \underbrace{p_C(c)}_{\text{Prior}}.$$

Since the denominator $p_X(x)$ is a normalization constant independent of c , the MAP rule is equivalently written as:

$$a^*(x) = a_{\underset{c \in \mathcal{C}}{\operatorname{argmax}}} p_{X|C}(x | c) p_C(c).$$

Numerical Example: 0–1 Loss in a 3-Class Problem

Suppose that for a given observation x the likelihood probabilities are:

$$p_{X|C}(x | 1) = 0.6, \quad p_{X|C}(x | 2) = 0.9, \quad p_{X|C}(x | 3) = 0.12.$$

and the prior for each class is $1/3$, i.e. $p_C(1) = p_C(2) = p_C(3) = \frac{1}{3}$. We then have $p_{X|C}(x | 1)p_C(1) = 0.2$, $p_{X|C}(x | 2)p_C(2) = 0.3$, and $p_{X|C}(x | 3)p_C(3) = 0.04$. The optimal decision is

$$a^*(x) = a_{\arg\max_{c \in \mathcal{C}} p_{X|C}(x | c)p_C(c)} = a_2,$$

which corresponds to choosing class 2.

5.5 Incorporating a Reject Option

In many critical applications (e.g., medical diagnosis), it is useful to include a *reject* option. Extend the action space to include a_{reject} and define its loss as

$$L(y, a_{\text{reject}}) = \lambda, \quad \text{with } 0 \leq \lambda < 1.$$

Then, the expected risks are:

- For action a_i :

$$R(a_i | x) = 1 - p_{C|X}(i | x).$$

- For the reject option:

$$R(a_{\text{reject}} | x) = \lambda.$$

The decision rule becomes

$$a^*(x) = \begin{cases} a_{\arg\max_{c \in \mathcal{C}} p_{C|X}(c | x)} & \text{if } \max_{c \in \mathcal{C}} p_{C|X}(c | x) > 1 - \lambda, \\ a_{\text{reject}} & \text{otherwise.} \end{cases}$$

Numerical Example: Reject Threshold in Binary Classification

Consider binary classification ($C \in \{0, 1\}$) with posterior $p_{C|X}(1 | x)$ and let $\lambda = 0.3$ (so the threshold is $1 - \lambda = 0.7$). Then:

- If $p_{C|X}(1 | x) > 0.7$, choose a_1 (class 1).
- If $p_{C|X}(1 | x) < 0.3$ (since $p_{C|X}(0 | x) = 1 - p_{C|X}(1 | x)$ would be > 0.7), choose a_0 (class 0).
- Otherwise, choose a_{reject} .

For instance, if $p_{C|X}(1 | x) = 0.65$, then neither class meets the threshold, so the optimal decision is to choose a_{reject} .

5.6 Example: Medical Testing

Consider a medical testing scenario:

- $C \in \{0, 1\}$, where $C = 0$ indicates “Disease is present” and $C = 1$ indicates “Disease is not present.”
- A measurement X (e.g., a biomarker level) is obtained with likelihood $p_{X|C}(x | c)$.
- The actions are $\mathcal{A} = \{\text{Treat}, \text{Do not treat}, \text{Reject}\}$.
- The loss function $L(y, a)$ is defined as:

$$L(c, a) = \begin{cases} \text{small} & \text{if } c = 0 \text{ and } a = \text{Treat}, \\ \text{very large} & \text{if } c = 0 \text{ and } a = \text{Do not treat}, \\ \text{moderate} & \text{if } c = 1 \text{ and } a = \text{Treat}, \\ \text{small} & \text{if } c = 1 \text{ and } a = \text{Do not treat}, \\ \lambda & \text{if } a = \text{Reject}. \end{cases}$$

The Bayesian approach is as follows:

1. Compute the posterior using

$$p_{C|X}(c | x) = \frac{p_{X|C}(x | c)p_C(c)}{p_X(x)}.$$

2. Calculate the expected loss $R(a | x)$ for each action.
3. Choose the action $a^*(x)$ that minimizes $R(a | x)$.

Numerical Example in Medical Testing

Assume:

- Disease prevalence: $p_C(0) = 0.01$ and $p_C(1) = 0.99$.
- Test sensitivity: $p_{X|C}(x | 0) = 0.90$.
- False positive rate: $p_{X|C}(x | 1) = 0.05$.

Then, the overall probability of a positive test is:

$$p_X(x) = 0.90 \cdot 0.01 + 0.05 \cdot 0.99 = 0.0585.$$

The posterior probability of disease given a positive test is:

$$p_{C|X}(0 | x) = \frac{0.90 \cdot 0.01}{0.0585} \approx 0.1538.$$

Even with a positive test, the low prior (base rate) of disease leads to a posterior of approximately 15.4%. Depending on the costs (losses) assigned, the decision might be to treat, not treat, or choose the reject option for further testing.